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SECTION-G
SUBJECT:SOFTWARE ENGG

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #63 — Community Solar Credit Allocation Manager (Sustainability & Green Tech)

1. Requirements Table

Exactly 5 Functional Requirements (FR-001 to FR-005) and 2 Nonfunctional Requirements (NFR-001, NFR-002). FR-001 and NFR-001 are the ones given in the problem statement; FR-002–FR-005 and NFR-002 complete the set.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record solar generation units (kWh) from smart meter feeds and allow prosumers to transfer surplus credits to neighbor accounts.	High	Pass: transferred credits deduct from the prosumer's balance and apply to the neighbor's billing statement. Fail: a transfer that exceeds the available balance is rejected.	Given in the official problem statement; core data-ingestion and P2P sharing capability.
FR-002	Functional	The system shall allow a Prosumer Resident to view their current solar credit balance and transfer history after authenticating.	High	After successful login, the balance shown matches the sum of recorded generation plus/minus completed transfers, and is refreshed within 3 seconds.	Prosumers need visibility into their available credits before deciding to transfer or rely on them for billing offset.
FR-003	Functional	The system shall validate that a peer-to-peer credit transfer does not exceed the sending prosumer's available balance before committing it.	High	For any transfer request, if amount > available balance the system rejects it and no ledger entry is created; if amount ≤ available balance both accounts are updated atomically.	Prevents overdraft of credits and keeps the microgrid ledger consistent.
FR-004	Functional	The system shall allow the Co-op Manager to generate a monthly billing offset statement for each prosumer showing generation, transfers, and net utility offset.	Medium	When a valid billing period is selected, the system produces one statement per prosumer containing generation totals, credits sent/received, and the resulting utility bill offset.	Co-op needs a documented, auditable basis for the monthly utility billing adjustment.
FR-005	Functional	The system shall flag a household's generation reading for Co-op Manager review when it deviates by more than a configurable threshold (e.g. ±20%) from that household's historical average.	Medium	When a new reading falls outside the configured threshold, a review flag is created and visible on the Co-op Manager dashboard within the same processing cycle.	Detects faulty smart meters or data-feed errors before they distort credit allocation.
NFR-001	Nonfunctional	The smart-meter telemetry processing pipeline must validate and store hourly generation data for 500 households with 99.9% uptime.	High	Pass: benchmarking/load tests confirm target latency and 99.9%	Given in the official problem statement; the
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				availability under simulated peak load.	ledger and billing offset are only trustworthy if ingestion is reliable.
NFR-002	Nonfunctional	The system shall encrypt all smart-meter telemetry and credit-transfer transactions in transit and at rest, and shall restrict billing-statement generation to authenticated Co-op Manager accounts.	High	Security test confirms telemetry and transfer data are encrypted (TLS in transit, AES-256 at rest); an unauthenticated or nonmanager user attempting to generate a billing statement is denied access.	Protects household energy data and prevents unauthorized changes to billing offsets.

2. Actors and Use Cases

Actor	Role
Prosumer Resident	A household with rooftop solar. Views their credit balance and transfers surplus credits to a neighbor within the microgrid.
Co-op Manager	Administers the neighborhood microgrid co-op: reviews flagged generation data, generates monthly billing offset statements, and applies manual adjustments when needed.
Smart Meter System	External device/telemetry feed that automatically reports each household's solar generation in kWh.
Utility Billing System	External system that receives the monthly credit/billing offset data so it can be applied to the household's utility bill.

2.1 Actors

2.2 Use Cases

ID	Use Case	Purpose
UC-01	**Ingest Smart Meter Generation Data**	Receive and store hourly solar generation readings for each household.
UC-02	**View Solar Credit Balance**	Let a prosumer see their current available credit balance and history.
UC-03	**Transfer Surplus Credits to Neighbor**	Let a prosumer send surplus credits to another household in the microgrid.
UC-04	**Validate & Calculate Credit Transfer**	Check the transfer amount against the available balance and compute the resulting balances. Always invoked by UC-03.
UC-05	**Generate Monthly Billing Offset Statement**	Produce each prosumer's monthly statement of generation, transfers, and utility bill offset.
UC-06	**Apply Manual Credit Adjustment**	Let the Co-op Manager optionally correct a statement, e.g. after resolving a flagged/anomalous reading.
UC-07	**Export Billing Offset to Utility System**	Send the finalized monthly offset data to the external utility billing system.

3. UML Relationships

Allocate/Transfer flow — «include»:

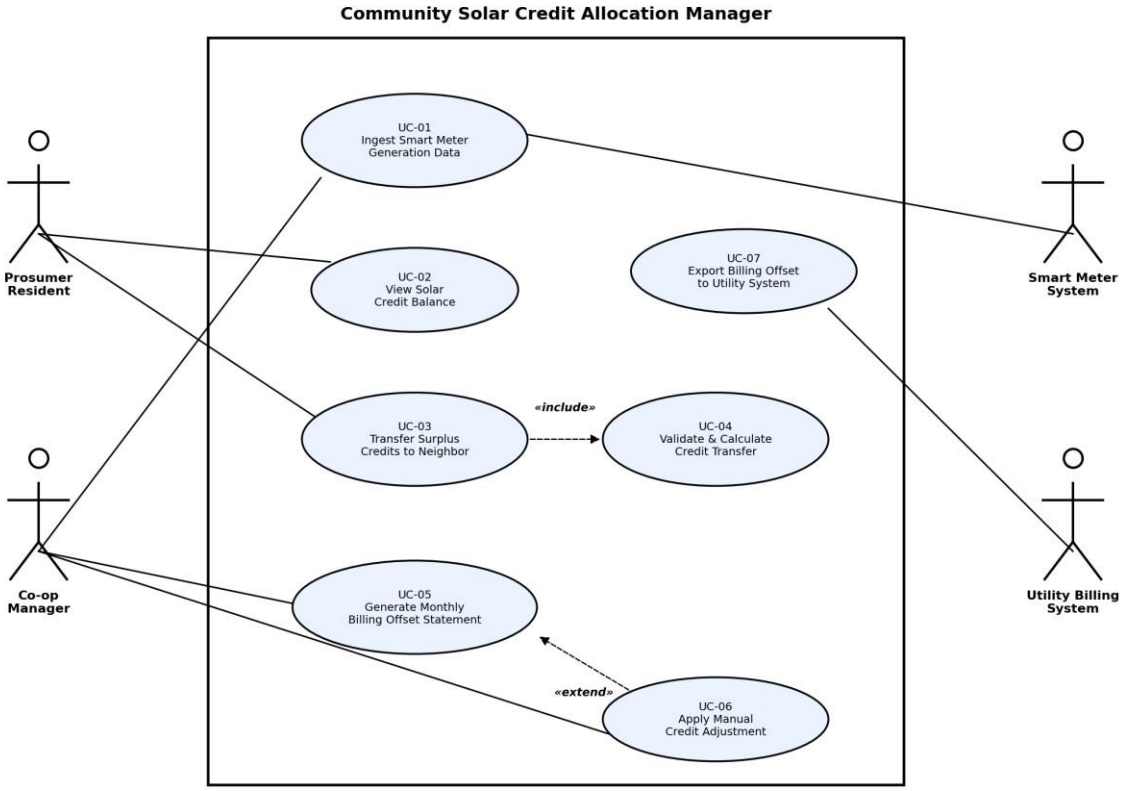
- UC-03 Transfer Surplus Credits to Neighbor «include» UC-04 Validate & Calculate Credit Transfer — every transfer must always validate the amount and calculate the resulting balances, so this behaviour is mandatory and factored out with include.

Billing statement flow — «extend»:

- UC-06 Apply Manual Credit Adjustment «extend» UC-05 Generate Monthly Billing Offset Statement — a manual adjustment is only added optionally, when the Co-op Manager decides a statement needs correcting (e.g. after a flagged anomalous reading), so it is modelled as extend, not include.

4. UML Use-Case Diagram

Actors are drawn outside the system boundary; all seven use cases are inside it; associations are solid lines; «include»/«extend» are dashed arrows.



5.Actor → Use Case Associations

Actor	Connects to (associations)
Prosumer Resident	UC-02 View Solar Credit Balance; UC-03 Transfer Surplus Credits to Neighbor
Co-op Manager	UC-01 Ingest Smart Meter Generation Data (oversight); UC-05 Generate Monthly Billing Offset Statement; UC-06 Apply Manual Credit Adjustment
Smart Meter System	UC-01 Ingest Smart Meter Generation Data
Utility Billing System	UC-07 Export Billing Offset to Utility System

6. Use-Case Flow Specification

Use Case: UC-03 — Transfer Surplus Credits to Neighbor

Primary Actor: Prosumer Resident Preconditions:

- The Prosumer Resident is registered and successfully authenticated.
- The prosumer has a recorded solar credit balance greater than zero (from ingested smart-meter generation data).
- The intended recipient is a valid neighbor account within the same microgrid/co-op.

Postconditions:

- Success: the transferred amount is deducted from the sender's balance and added to the neighbor's balance; both accounts' transaction histories are updated and available for the next monthly billing offset statement.
- Failure: no balance is changed and no transaction record is created.

Main Success Scenario:

- 1. Prosumer Resident logs into the portal.
- 2. System authenticates the prosumer.
- 3. Prosumer selects “Transfer Surplus Credits”.
- 4. System displays the prosumer's current credit balance and the list of eligible neighbor accounts in the microgrid.
- 5. Prosumer selects a neighbor and enters the credit amount to transfer.
- 6. System validates and calculates the transfer («include» UC-04): checks that the amount does not exceed the available balance.
- 7. System deducts the transferred credits from the prosumer's balance.
- 8. System credits the neighbor's account.
- 9. System records the transaction in both accounts' history.
- 10. System displays a transfer confirmation to the prosumer.
- 11. Use case ends successfully.

Alternate Flow — Insufficient Balance:

- 6a. System detects that the requested transfer amount exceeds the prosumer's available balance.
- 6a1. System displays an “insufficient balance” error and does not create any transaction.
- 6a2. Prosumer either lowers the amount or cancels the transfer.
- 6a3. If the prosumer resubmits with a corrected amount, the flow returns to step 6.